

Physics-Informed Weakly-Supervised Learning for Quality Prediction of Manufacturing Processes

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Abstract—In manufacturing processes, a multitude of sensors are typically deployed to collect data of process parameters. While this provides an opportunity to better predict and control the quality of the final product, the relationship between the process variables and the desired final quality is often not well-understood. To establish this relationship, machine learning (ML) models can be used. However, collecting large, labeled datasets to train the ML model can be difficult, as creating such datasets typically involves costly or destructive end-of-line quality testing of the products. To overcome this challenge, we propose a novel framework called Physics-informed Weakly-supervised Learning (PWL) that integrates physics-based models with data-driven ML models. By leveraging physical knowledge and using the outputs of physics-based models as weak labels, PWL offers an alternative to traditional methods that require large, labeled datasets. Our approach simultaneously optimizes the data-driven ML model, as well as the discrepancy and calibration parameters of the physics-based model, resulting in superior predictive performance compared to either model used alone. We demonstrate the effectiveness of PWL through simulation experiments, comparisons with existing methods, and two real-world case studies, highlighting its potential for improving quality prediction in various manufacturing systems.

Note to Practitioners—The proposed method in this paper, Physics-informed Weakly-supervised Learning (PWL), addresses a common problem in manufacturing processes, where obtaining large, labeled datasets can be costly or not always possible. By integrating physics-based models with data-driven ML models, PWL leverages available physical knowledge to improve the predictive performance of product quality in manufacturing systems. This enables practitioners to better understand and optimize their manufacturing processes even with limited labeled datasets, leading to improved product quality and reduced production costs.

Index Terms—Weakly-supervised learning, physics-informed machine learning, manufacturing processes, quality prediction.

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I. INTRODUCTION

IN MANUFACTURING processes, a multitude of sensors are deployed to collect data of process parameters. This provides an opportunity to better predict and control the quality of the final product. However, the relationship between the process variables and the desired final quality is often not well understood. To establish this relationship, machine learning (ML) models can be built based on historical datasets which contain the final quality of each product (Y) and the process variables involved in making the product (X), known as labeled datasets. However, it can be challenging to obtain the labels of products, which often involve comprehensive end-of-line quality testing. This can be costly and often infeasible for all products in the case of destructive testing techniques. As a result, the labeled samples used to train ML models are limited. This challenging situation is intensified by the fact that industrial processes are inherently noisy. Without sufficient data, it would be difficult to build robust models and consequently model-based quality prediction would have poor performance.

Recently, there has been some research interest in integrating physics-based models with data-driven ML models to improve the predictive performance [7], [11], [12], [18], [25], [26], [32], [33], [37], [41], [43], [47], [49], [55]. Physics-based models exploit known information from physical knowledge to understand the behavior of a manufacturing process. Such models are commonly implemented using either mathematical equations or computer simulation, both based on physical laws and principles. These models are often more cost-effective than obtaining labeled samples via end-of-line quality testing.

Here we provide a motivating example for a spot-welding process shown in Fig. 1. Destructive testing is still the most common method to test spot weld properties with the two most common testing methods being peel and chisel tests [52]. However, these methods are time-consuming, costly, and infeasible in continuous production. To avoid destroying samples, physics-based models have been used to predict final quality features related to the spot-welding process. For instance, consider the case that the final quality we are interested in predicting is temperature difference of the weld [ΔT]. Process variables in the welding process include force [kN], current [kA], time of weld [s], and mass [g]. A physics-based model will utilize mathematical formula from physics to produce a prediction for the final quality, temperature difference. The input of the mathematical formula includes some of the

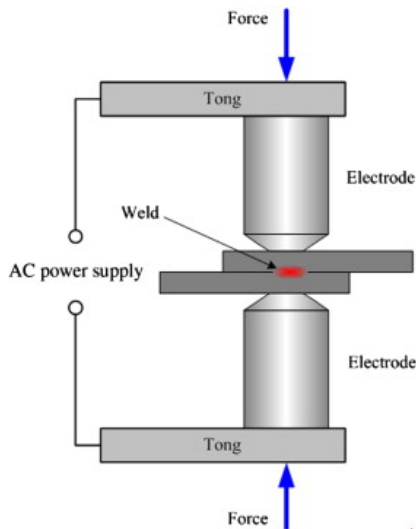


Fig. 1. Spot welding process.

process variables such as current, time of weld, and mass, as well as parameters related to fundamental mechanical principles such as elastic-plastic part deformation and thermal expansion [56]. The output from the mathematical formula is an estimate of the true temperature difference of the weld $[\Delta T]$.

Physics-based models can be utilized to provide an estimate of the final quality of the product, which will mitigate the shortage of labeled samples for building data-driven ML models. However, physics-based models have limitations. First, it is typical that physics-based models can only take a small number of process variables as input. Thus, there may be a discrepancy between the output of physics-based models and the product quality. Also, physics-based models need to include some parameters related to the physical principles that are not observable. To estimate the unobservable parameters, a technique referred to as model calibration is needed [35]. However, the calibration process may be inadequate because it often relies on oversimplified models and flawed assumptions, which can lead to inaccurate predictions [45]. Hence, a physics-based model can be inexact in characterizing the process-quality relationship, and thus may not be used as a stand-alone model.

Integrating physics-based models with data-driven ML models presents several challenges:

- 1) Uncertainty in final quality estimates: Physics-based models are often subject to various sources of uncertainty, requiring model discrepancy to be addressed before using the model.
- 2) Indirect measurement of final quality: The output of a physics-based model may be highly correlated with the final quality but may not directly measure it in some cases.
- 3) Calibration issues: Physics-based models need to be calibrated, but calibration is often performed using ad-hoc procedures rather than optimization.

Several studies in the literature have attempted to incorporate physics knowledge into their frameworks, yet none have

fully tackled the challenges mentioned earlier. Some involve integrating physics knowledge into deep learning models [11], [12], [18], [25], [32], [55], many of which find applications in additive manufacturing [25], [32], [55]. For instance, one method leverages dimensionality augmentation to incorporate information derived from the selective laser melting process into the model [55]. Other strategies for incorporating physics knowledge into models include the use of physically-informed priors [18], incorporating physics constraints within the loss function [32], utilizing outputs from physics models as additional inputs [32], and pre-training with input-output data from physics models [32]. Despite the innovative approaches, a pervasive issue remains: these strategies do not directly address data scarcity, a significant concern given the data-intensive nature of deep learning models [11], [12], [18], [25], [32], [55]. Furthermore, they overlook the calibration challenges associated with physics models. While some research has attempted to tackle the calibration issue through a two-stage approach [11], it remains isolated and not fully integrated with the overall prediction process.

Another area of research involves integrating physics knowledge through semi-supervised learning techniques. One common approach is utilizing large databases of unlabeled physics events and focusing on uncovering data patterns and correlations to facilitate label spreading [41]. Another approach focuses on utilizing simulations based on established physical theories and principles to create a dataset that improves training [7], [26], [37]. For example, Herrero-Garcia et al. [26] pre-trained their model using simulations from established physics theories to represent known physical phenomena in dark matter, then trained on real experimental data. Moreover, Chakravarti et al. [7] presented a method that leverages a known physics model to generate a background dataset and then uses a classifier to differentiate between this background data and the experimental data. Semi-supervised learning methodology shows promising results in addressing data scarcity issues; however, these approaches heavily rely on the quality and representativeness of the simulated data from the physical model. Moreover, such approaches do not address the challenges central to physics-based models, such as uncertainty in final estimates and calibration issues.

To address the challenges, we draw upon ideas from the field of weakly-supervised learning in ML [57]. Weakly-supervised learning deals with scenarios where it is difficult to obtain a large number of samples with precise labels to train an ML model, but it is relatively easy to obtain unlabeled samples to provide weak supervision for the ML model training. We consider the output from physics-based models as providing weak labels for samples. Driven by this idea, we propose a novel framework for Physics-informed Weakly-supervised Learning (PWL) in this paper. The novelties of PWL are summarized as follows:

- 1) PWL utilizes a data-driven approach to handle the uncertainty and inadequacy of physics-based models.
- 2) PWL utilizes the outputs of physics-based model as weak labels, and integrates them with limited labeled data collected from the manufacturing process.

- 3) PWL simultaneously optimizes the data-driven ML model as well as the discrepancy and calibration parameters of the physics-based model, thus achieving a better performance than each type of model used alone.

The rest of this paper is structured as follows: Section II discusses related research and highlights the gaps in existing work, providing motivation for the development of PWL. Section III presents the model, algorithm, and theoretical property of PWL. Section IV presents simulation experiments and comparison of PWL with existing methods. Section V provides two real-data case studies of manufacturing processes to demonstrate the real-world applications of PWL. Finally, Section VI provides the conclusion.

II. RELATED WORK

The model we are proposing has significant connections with two established research fields: physics-informed ML and weakly-supervised learning. This section will present a summary of both research fields and highlight the unique aspects of our work.

A. Physics-Informed ML

This research field aims to integrate physical laws, principles, and models with data-driven ML [51]. Due to the variety of forms in which physics is represented in various application domains, a diverse collection of methods is needed to develop physics-informed ML models. These methods can be generally divided into two categories: i) physics-informed architecture ii) physics-informed regularization [40].

Physics-informed architecture is a recent research direction that can make use of specific characteristics of the problem being solved to implicitly encode physical consistency or other desired physical properties in designing the ML model. This area is focused largely on neural network architecture, due to their modular and flexible nature that makes them a prime candidate for architecture modification. For example, in lake temperature modeling, Daw et al. [15] incorporated a physical intermediate variable as part of a monotonicity-preserving structure. Similarly, in the paper by Muralidhar et al. [42], physics-constrained variables were inserted as the intermediate variables in the convolutional neural network (CNN) architecture, which achieved significant improvement in predicting drag force on particle suspensions in moving fluids. Other research focuses on designing computational graphs that mimic the behavior of physics-based methods in preserving energy conservation laws [13], [24], [39]. Despite the promising features of physics-informed architecture, formulating a trainable architecture that represents the problem at hand is a challenging task [25]. Moreover, even if an architecture is successfully designed, it is often highly application-specific. Hence, a new architecture needs to be formulated and trained for any change in the initial application conditions.

Physics-informed regularization could integrate physical understanding into the loss function of an ML model, which helps generate results that are consistent with established physical laws [30], [34], [48]. Pure data-driven ML can fail to produce such results directly from data, especially when

provided with only limited samples. For instance, in the context of lake temperature level modelling, one approach was to include an extra physics-based penalty that guaranteed predictions of denser water were at reduced depths than those of much less thick water, which was a known monotonic relationship [14]. Moreover, other approaches may attempt to create a regularization term that accounted for the underlying physics by penalizing divergence from the governing equations, such as Burgers' equation [47]. However, physics-informed regularization may not be feasible due to several challenges [33]:

- 1) Mismatched input variables: Physics-informed regularization may not be feasible if the physics-based model and the ML model do not share the same input variables.
- 2) Limited generalizability: Regularization with limited samples may restrict the model's generalizability, leading to performance deterioration.
- 3) Incomplete or imprecise knowledge: It is essential to recognize that the knowledge from physics-based models may be incomplete or imprecise.
- 4) Complexity of large-scale neural networks: Physics-informed ML models often involve training large-scale neural networks with complicated loss functions, which can be highly non-convex optimization problems.
- 5) Competing loss function terms: The terms in the loss function may compete with each other during training, making the training process less robust and stable, and potentially preventing convergence to the global minimum.

An integration framework that is capable of integrating incomplete/imprecise physical knowledge with ML is highly desirable, which is the goal of this research.

B. Weakly-Supervised Learning

In ML, the models trained using only labelled samples are known as supervised learning models. In many applications, there may be samples that are not precisely labelled but can still be useful to provide some "weak" supervision for the ML model. This field is known as weakly-supervised learning. Research in weakly-supervised learning can be categorized based on the quality and characteristics of weakly labeled data to three main fields: inexact supervision, inaccurate supervision, and incomplete supervision [57]. Inexact supervision refers to situations where the labels or annotations provided are incomplete or noisy, meaning that they do not fully capture the desired concept or class [16], [21]. For example, in image classification, an image might have multiple objects, but only one object is labelled [10]. Inaccurate supervision, on the other hand, refers to situations where the provided labels or annotations are wrong. For example, an image of a dog might be labeled as a cat [22]. In the case of incomplete supervision, a portion of the training samples is labeled, with the remainder being unlabeled. This presents unique challenges as the algorithm must learn from both labeled and unlabeled data, often requiring innovative approaches to effectively leverage the unlabeled portion. Techniques like semi-supervised learning [8], [59] and active learning [50] are

commonly employed under incomplete supervision, aiming to maximize learning efficiency from limited labeled data. Active learning operates under the assumption of having an ‘oracle’, like a human expert, who can provide labels for chosen unlabeled samples. The objective of active learning methods is to identify the most informative unlabeled samples to send them to the ‘oracle’ for querying.

In the semi-supervised learning (SSL) approaches, the dependency on an external ‘oracle’ for guidance is eliminated. This approach combines the use of both labeled and unlabeled data to enhance model training in areas such as classification and prediction. Several SSL algorithms exist, with some based on the concept that data naturally forms clusters, suggesting that samples in the same cluster share similar class labels [8]. Graph-based learning, a key technique in SSL, involves creating graphs where nodes are training samples and edges reflect their interrelations [4], [23]. Other strategies consider the labels for unlabeled data as simply missing, operating under the premise that all data, labeled or not, are derived from a unified model [44]. Furthermore, certain SSL methods focus on pinpointing boundaries in less densely populated data areas while ensuring the accurate classification of labeled data [9], [31], [38]. However, the field of SSL, while extensive, has yet to fully explore incorporating physics-related information from unlabeled samples into model building.

Moreover, recent work in weak supervision capitalizes on domain knowledge to facilitate ranking unlabeled data samples, a strategy that has found considerable applicability in healthcare settings [19]. Particularly, this methodology [19] hinges on utilizing expert insights to generate relative rankings based on anticipated reactions to treatments, such as medication effects in Parkinson’s disease management, thus providing a binary label for a pair of unlabeled samples. However, this strategy finds limited applicability in the manufacturing sector because it lacks a parallel to the healthcare knowledge-driven ranked samples. Also, the manufacturing domain, governed rigidly by physical laws, demands a more structured and quantifiable approach to data labeling derived from established physical principles.

C. Gaps

The proposed PWL model integrates the output of the physics-based model with data-driven ML, while at the same time optimizing the physics-based model by estimating the discrepancy and calibration parameters of the physics-based model. In this way, the integration can be more effective. In comparison, the existing works in physics-informed ML integrate physical laws, principles, and models as they are, without optimizing them. In the field of weakly-supervised learning, none of the existing works consider weak supervision from the physics-based model. In this sense, PWL is also substantially different from the existing weakly-supervised models in ML.

III. PHYSICS INFORMED WEAK LEARNING (PWL)

In this section, we will present the development of PWL. Section III-A defines the problem setting and notations.

Section III-B proposes the model formulation. Section III-C proposes an optimization problem for model estimation. Section III-D presents the algorithm for solving the optimization and the associated theoretical properties.

A. Notations and Problem Statement

Let $\mathcal{Y}$ denote the output variable (i.e., product quality) of a manufacturing process. $\mathcal{Y}$ is typically affected by a set of variables and parameters that are responsible for making the final product. In practice, we are interested in predicting the product quality $\mathcal{Y}$ either empirically using a physics-based model or a data-driven ML model.

1) *Physics-Based Models and Limitations:* Physics-based models depend on physical understanding of the system’s input and output, and characterize the relationship between the input variables and the output variable as follows:

$$\mathcal{Y}_i = \varphi[\eta(\mathbf{x}_i^{ph}, \boldsymbol{\theta})] + \varepsilon_i^{disc} + \varepsilon_i, \quad (1)$$

where $\eta(\cdot, \cdot) : \mathbb{R}^{(q_1+q_3)} \rightarrow \mathbb{R}^z$ is the deterministic output from a physics-based model, φ is a mapping function that maps from the physics-based model output to the product quality, $\varepsilon_i^{disc} \in \mathbb{R}$ is an unknown model discrepancy term between the mapped physics-based model output and the product quality, $\mathbf{x}_i^{ph} \in \mathbb{R}^{q_1}$ are input variables to the physics-based model, $\boldsymbol{\theta} \in \mathbb{R}^{q_3}$ are calibration parameters which cannot be directly controlled or measured in a real manufacturing process, such as material properties, boundary conditions, etc., $\varepsilon_i \in \mathbb{R}$ is a random error, and i is the index for samples such as products made by the manufacturing process.

Generally, physics-based models are a complex mathematical representation of the true system that are implemented through a computer model or a physical equation. The mathematical representation may not accurately emulate the process observations mainly due to two reasons:

First, application of physics-based models is accompanied by various sources of uncertainty. In our framework, we address two sources of uncertainty that significantly affect the performance of a physics-based model. The first source is commonly referred to as model discrepancy. Model discrepancy can be defined as the difference between the output of the real-world manufacturing process such as product quality and the output of the physics-based model at the same inputs’ values. Hence, even if we know the true values of all the inputs required to generate an output from the physics-based model, this output may not be equal to the true output of the manufacturing process, thus creating a discrepancy denoted by ε_i^{disc} in Eq. 1. This discrepancy exists because of insufficient inputs. Physics-based models are built only on input variables which have a well-understood relationship with the model’s output based on physical knowledge. Hence, it is likely that many variables in the manufacturing process cannot be included in the physics-based models.

Second, physics-based models have unknown parameters that need to be calibrated, denoted by $\boldsymbol{\theta}$ in Eq. 1. Calibration can be defined as the activity of adjusting $\boldsymbol{\theta}$ until the output of the physics-based model fits the observed data from the real manufacturing process [35]. In practice, calibration is

typically used as an ad-hoc fitting in a pre-step. The fitted calibration parameter is substituted into the physics-based model to generate an output. This sequential approach may be sub-optimal due to inefficiencies in the two-step process, potential overfitting during ad-hoc fitting, incomplete parameter optimization, and a lack of joint optimization between the physics-based and ML models. An alternative approach that integrates both models more effectively can lead to better predictive performance and improved understanding of the manufacturing process [47].

2) *Data-Driven ML Models and Limitations:* Due to the advanced sensing capability of modern manufacturing systems, $\mathbf{x}^{ph}$ are not the only measurable variables in the system. Hence, we define another set of variables $\mathbf{x}^{pr} \in \mathbb{R}^{q_2}$ which represent measurable variables in a real system that are not included in the physics-based model. This motivates another modeling approach, data-driven models, to model the relationship between all measured variables in the system and $\mathcal{Y}$ as follows:

$$\mathcal{Y}_i = f(\mathbf{x}_i^{ph}, \mathbf{x}_i^{pr}) + \varepsilon_i, \quad (2)$$

where $f: \mathbb{R}^{(q_1+q_2)} \rightarrow \mathbb{R}$ is an ML model to be trained based on observational data.

In practice, the performance of data-driven models is hindered by two main challenges. Firstly, the number of labeled samples can be very limited, making the model training less robust. Secondly, these models follow a pure data-driven approach without taking physical knowledge into consideration. In fact, we may consider physical knowledge to provide a form of “virtual/synthetic data” to help mitigate the shortage of labeled samples.

3) *A Joint Model Perspective:* We seek to integrate physics-based models and data-driven models in a unified framework to leverage their complementary power. To further highlight the significance of this approach, let us revisit our motivating example, the spot-welding process: The product quality variable y is weld temperature difference. The physics-based model $\eta(\cdot, \cdot)$ is a mathematical equation that utilizes principles of thermal resistivity and conductivity. The inputs to the physics-based model, $\mathbf{x}^{ph}$, include the current magnitude and time of the weld. The unknown calibration parameters θ in the physics-based model include thermal conductivity and resistivity coefficients. On the other hand, there are other variables in the process that are not included in $\mathbf{x}^{ph}$, i.e., $\mathbf{x}^{pr}$ which includes the force magnitude. A data-driven model can be built based on $\mathbf{x}^{ph}$ and $\mathbf{x}^{pr}$. It can be seen that neither model is sufficient by itself to produce an adequate predictive performance, because they neglect the fact that $\mathcal{Y}$ is jointly influenced by the three sets of variables as follows:

$$\mathcal{Y}_i = u(\mathbf{x}_i^{ph}, \mathbf{x}_i^{pr}, \theta) + \varepsilon_i, \quad (3)$$

where $u(\cdot, \cdot, \cdot) : \mathbb{R}^{(q_1+q_2+q_3)} \rightarrow \mathbb{R}$ is the underlying unknown true process-quality relationship.

Models that are developed without taking Eq. 3 into consideration will lead to suboptimal solutions. PWL focuses mainly on the improvement and integration of the two modeling approaches, physics-based models and data-driven models, while respecting the assumption in Eq. 3.

B. Joint Model Formulation

It is difficult to directly formulate a joint model as in Eq. 3. We propose the following formulation:

$$\mathcal{Y}_i = \mathbf{H}(\mathbf{x}_i^{ph}, \theta) \mathbf{g} + \mathbf{B}(\mathbf{x}_i^{pr}, \mathbf{x}_i^{ph}) \mathbf{d} + \varepsilon_i, \quad (4)$$

where $\mathbf{H} : \mathbb{R}^{q_1+q_3} \rightarrow \mathbb{R}^p$ are basis functions for creating patterns associated with $\mathbf{x}_i^{ph}$ and θ , $\mathbf{g} \in \mathbb{R}^p$ are coefficients vectors associated with the patterns created by $\mathbf{H}$; $\mathbf{B} : \mathbb{R}^{q_1+q_2} \rightarrow \mathbb{R}^b$ are basis functions for creating patterns associated with $\mathbf{x}_i^{pr}$ and $\mathbf{x}_i^{ph}$, $\mathbf{d} \in \mathbb{R}^b$ are coefficients vectors associated with the patterns created by $\mathbf{B}$. The basis matrix formed by $\mathbf{H}$ and $\mathbf{B}$ are used to represent patterns in the original feature matrix, and the coefficients $\mathbf{g}$ and $\mathbf{d}$ are estimated based on the patterns in $\mathbf{H}$ and $\mathbf{B}$, respectively. By using basis, the objective function aims to identify the most important patterns in the data for predicting the response variable such as product quality.

The structure of the model proposed in Eq. 4 offers three significant advantages. First, the model is structured in a way that it allows the first term to have the same input as the physics-based model to allow knowledge distillation. Knowledge distillation [29] is a technique used in ML to transfer knowledge from a larger, more complex model (known as the teacher model) to a smaller, simpler model (known as the student model). This will help address labeled data shortage by allowing the model to additionally learn from the abundant weakly labeled samples produced by the physics-based model $\eta(\cdot, \cdot)$. Second, the second term helps mitigate the discrepancy of using the physics-based model alone. Since this term is built upon both $\mathbf{x}^{ph}$ and $\mathbf{x}^{pr}$, it can leverage the data-driven model. Overall, the model in Eq. 4 combines physics-based and data-driven models together. Third, this is a highly flexible framework that can accommodate various types of basis functions. By choosing different basis functions, the model can be tailored to specific problem settings, allowing it to capture diverse patterns and relationships in the data. This adaptability makes the proposed framework suitable for a wide range of applications in manufacturing and other domains.

Notably, in the model in Eq. 4, the calibration parameters θ are integrated with the model formulation, which makes it possible to optimize θ together with other model parameters, instead of making calibration a pre-step.

C. Optimization

In order to estimate the model parameters and calibration parameters for the model in Eq. 4, we propose the following optimization:

$$\begin{aligned} \min_{\mathbf{g}, \theta, \mathbf{d}} & \sum_{i=1}^n \left\| \mathcal{Y}_i - \mathbf{H}(\mathbf{x}_i^{ph}, \theta) \mathbf{g} - \mathbf{B}(\mathbf{x}_i^{pr}, \mathbf{x}_i^{ph}) \mathbf{d} \right\|_F^2 \\ & + \lambda_1 \sum_{j=1}^N \left\| \mathbf{H}(\mathbf{x}_j^{ph}, \theta) \mathbf{g} - \varphi[\eta(\mathbf{x}_j^{ph}, \theta)] \right\|_F^2 \\ & + \lambda_2 \|\mathbf{g}\|_1 + \lambda_3 \sum_{q=1}^{(q_1+q_3)} \|\mathbf{g}_q\|_2 \\ \text{s.t. } & \omega_l \leq \theta \leq \omega_u, \end{aligned} \quad (5)$$

where $\omega_l \in \mathbb{R}^{q_3}$ and $\omega_u \in \mathbb{R}^{q_3}$ are lower and upper bounds of the calibration parameters θ . N is the number of samples

generated by the physics-based model, while n is the number of labeled samples collected from the manufacturing system. N is generally much larger than n . Next, we explain the purpose of each term in Eq. 5:

- 1) The first term comes from Eq. 4, which aims to minimize the prediction error on the product quality variable, which is the overall goal of the framework.
- 2) The second term has two purposes. The first purpose is to regularize how much the $H(x_i^{ph}, \theta)g$ learns from the physics-based model $\eta(\cdot, \cdot)$. The amount of regularization is dependent on the accuracy of the physics-based model. If the physics-based model is inaccurate, regularization should be relatively weak, which can be realized by a smaller λ_1 . The second purpose is to handle the mismatch between the quality variable y_i and the physics-based model output $\eta(\cdot, \cdot)$. That is, they can be highly correlated but not necessarily measure the same thing. Hence, we introduced the φ function.
- 3) The third term is the L_1 penalty, which encourages sparsity at the individual coefficient level by applying an L_1 norm penalty to all the coefficients in g . This penalty term helps to identify the most important individual variables for predicting the outcome.
- 4) The fourth term is a group lasso penalty, which encourages group sparsity by applying an L_2 norm penalty to each group of coefficients g_q separately and then summing over all the groups. This penalty term helps to identify the most important groups of variables for predicting the outcome.
- 5) Together, the third and fourth terms form a combination of group selection and variable selection methods, known as the sparse group lasso. The parameters λ_2 and λ_3 control the trade-off between group sparsity and individual sparsity, respectively. This is required since we are generating patterns related to the x_i^{ph} and θ using the basis function H . Some of those patterns are related to the same phenomena, and hence we desire their effects to be chosen or unchosen together. Moreover, the patterns generated by H are not necessarily independent, and hence this penalty can help with identifiability issues in our model which will be discussed in more details in Section III-D3.

The optimization model in Eq. 5 jointly optimizes the physics-based model and data-driven model to achieve better predictive performance. Also, the model optimizes the calibration parameters θ in the physics-based model in the integrated framework. The proposed model in Eq. 5 is clearly non-convex and highly complicated to optimize. The algorithms developed to optimize the model will be discussed in detail in Section III-D.

D. Algorithm

The problem in Eq. 5 is not convex. However, it can be solved through a Blockwise Coordinate Descent (BCD) algorithm, where we iteratively update the g, θ , and d as shown in Algorithm 1.

Algorithm 1 The PWL algorithm

Input: $H(\cdot, \cdot), B(\cdot, \cdot), \varphi(\cdot), \varepsilon_0$
Initialize: $d = d^0, \theta = \theta^0$
While ($\|y_i - (H(x_i^{ph}, \theta)g - B(x_i^{pr}, x_i^{ph})d)\|^2 \geq \varepsilon_0$)
 1) Update g According to ADMM Algorithm (Section III-D1.)
 2) Update d as:
 $\min_d \sum_{i=1}^n \|y_i - H(x_i^{ph}, \theta)g - B(x_i^{pr}, x_i^{ph})d\|_F^2$
 3) Update θ according to ADMM algorithm (Section III-D2.)
End

Even though the problem in Eq. 5 is not convex, the three updates in Algorithm 1 are convex. In the PWL algorithm, there are two intuitive considerations about the initialization. First, we want to initialize d using small values to allow g to learn from the physics-based model in the first update. Second, we want to initialize θ to give the best possible performance of the physics-based model $\eta(\cdot, \cdot)$; hence it will be initialized according to our physics-based model calibration approach that to be discussed in Section IV. The d update has a close-form solution since we can cast it as a least square problem. On the other hand, the g and θ updates are not as simple and will be discussed in detail in Sections III-D1. and III-D2, respectively. Lastly, we will discuss properties of our algorithm related to identifiability and uniqueness in Section III-D3.

1) *The g Update:* In the g update, the problem in Eq. 5 reduces to solving the following problem:

$$\begin{aligned} \min_g \sum_{i=1}^n & \|y_i - H(x_i^{ph}, \theta)g - B(x_i^{pr}, x_i^{ph})d\|_F^2 \\ & + \lambda_1 \sum_{j=1}^N \|H(x_j^{ph}, \theta)g - \varphi[\eta(x_j^{ph}, \theta)]\|_F^2 \\ & + \lambda_2 \|g\|_1 + \lambda_3 \sum_{q=1}^{(q_1+q_3)} \|g_q\|_2. \end{aligned} \quad (6)$$

The problem in Eq. 6 is a sum of convex functions and hence convex. Assuming the basis functions $H(x_i^{ph}, \theta), B(x_i^{pr}, x_i^{ph})$ and $H(x_i^{ph}, \theta)$ were selected, to facilitate the estimation procedure, we represent them in the matrix form as $H_1 \in \mathbb{R}^{n \times p}$, $B_1 \in \mathbb{R}^{n \times b}$ and $H_2 \in \mathbb{R}^{N \times p}$, respectively. The problem in Eq. 6 is reformulated as follows:

$$\min_z \sum_{m=1}^M f_m(z), \quad (7)$$

where

$$\begin{aligned} M &= 4, f_1(g) = \|Y - H_1 g - B_1 d\|_F^2, f_2(g) \\ &= \lambda_1 \|H_2 g - \tilde{\eta}\|_F^2, f_3(g) \\ &= \lambda_2 \|g\|_1, f_4(g) = \lambda_3 \sum_{q=1}^{(q_1+q_3)} \|g_q\|_2, \tilde{\eta} \\ &= \begin{bmatrix} \varphi[\eta(x_1^{ph}, \theta)] \\ \vdots \\ \varphi[\eta(x_N^{ph}, \theta)] \end{bmatrix}, Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}. \end{aligned}$$

An effective and efficient algorithm to solve the problem in Eq. 7 can be developed using the Alternating Direction Method

Algorithm 2 The ADMM consensus algorithm

Input: step size Δ, ϵ
Initialize: replicates $\mathbf{z}_m^{(0)} = \mathbf{0}, \mathbf{u}_m^{(0)} = \mathbf{0}, \bar{\mathbf{z}} = \mathbf{0}$, for $m = 1, \dots, M$, of the same shape as $\mathbf{z}$.
For $k = 1, 2, \dots$ **do**
 Update $\mathbf{z}_m^{(k)} = \text{prox}_{\Delta f_m} \left[\bar{\mathbf{z}}^{(k-1)} - \mathbf{u}_m^{(k-1)} \right]$, for
 $m = 1, \dots, M$, in parallel.
 $\bar{\mathbf{z}}^{(k)} = \sum_{m=1}^M \frac{\mathbf{z}_m^{(k)}}{M}$
 $\mathbf{u}_m^{(k)} = \mathbf{u}_m^{(k-1)} + \mathbf{z}_m^{(k)} - \bar{\mathbf{z}}^{(k)}$
 If $\left| \bar{\mathbf{z}}^{(k)} - \bar{\mathbf{z}}^{(k-1)} \right| < \epsilon$
 Stop
 End
End

of Multipliers (ADMM) consensus algorithm [46], summarized in Algorithm 2 below. In order to apply the ADMM consensus algorithm to this problem, we first derive the proximal operator for the objective function, which involves computing the optimal solution for each variable while keeping the other variables fixed. The proximal operator is a key component in the ADMM framework and is crucial for the convergence and stability of the algorithm. By incorporating the proximal operator into the ADMM consensus algorithm, we can efficiently solve the optimization problem in Eq. 7 while ensuring a robust and stable solution.

Even To implement Algorithm 2 in solving Eq. 7, we need to evaluate the proximal operators of $\Delta f_1, \dots, \Delta f_M$. The results are in Proposition 1, and the derivation is in Section A. in in the supplementary materials.

Proposition 1: Let $f_1(\mathbf{g}) = \|\mathbf{Y} - \mathbf{H}_1 \mathbf{g} - \mathbf{B}_1 \mathbf{d}\|_F^2$, $f_2(\mathbf{g}) = \lambda_1 \|\mathbf{H}_2 \mathbf{g} - \tilde{\eta}\|_F^2$, $f_3(\mathbf{g}) = \lambda_2 \|\mathbf{g}\|_1$, and $f_4(\mathbf{g}) = \lambda_3 \sum_{q=1}^{(q_1+q_3)} \|\mathbf{g}_q\|_2$. The proximal operators of $\Delta f_1, \dots, \Delta f_4$ are given as follows: Let $\mathbf{z}$ and $\mathbf{a}$ have the same size as $\mathbf{g}$.

- 1) If $\mathbf{z} = \text{prox}_{\Delta f_1}[\mathbf{a}]$, we have
 $\text{prox}_{\Delta f_1}[\mathbf{a}] = (\mathbf{I} + \Delta \mathbf{H}_1^T \mathbf{H}_1)^{-1} (\mathbf{a} + \Delta \mathbf{H}_1^T (\mathbf{Y} - \mathbf{B}_1 \mathbf{d}))$,
where $\mathbf{I}$ is the identity matrix of order p .
- 2) If $\mathbf{z} = \text{prox}_{\Delta f_2}[\mathbf{a}]$, we have
 $\text{prox}_{\Delta f_2}[\mathbf{a}] = (\mathbf{I} + \lambda_1 \Delta \mathbf{H}_2^T \mathbf{H}_2)^{-1} (\mathbf{a} + \lambda_1 \Delta \mathbf{H}_2^T \tilde{\eta})$,
where $\mathbf{I}$ is the identity matrix of order p .
- 3) If $\mathbf{z} = \text{prox}_{\Delta f_3}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta f_3}[\mathbf{a}] = S_{\lambda_2 \Delta}(\mathbf{a}) = \begin{cases} \mathbf{a} - \lambda_2 \Delta, & \mathbf{a} > \lambda_2 \Delta \\ \mathbf{0}, & -\lambda_2 \Delta \leq \mathbf{a} \leq \lambda_2 \Delta \\ \mathbf{a} + \lambda_2 \Delta, & \mathbf{a} \leq -\lambda_2 \Delta. \end{cases}$$

- 4) If $\mathbf{z} = \text{prox}_{\Delta f_4}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta f_4}[\mathbf{a}] = \left[\mathbf{1} - \frac{\lambda_3 \Delta}{\|\mathbf{a}\|_2} \right]_+ \mathbf{a}.$$

2) *The θ Update:* In the θ update, the problem in Eq. 5 reduces to solving the following problem:

$$\begin{aligned} \min_{\theta} \sum_{i=1}^n \left\| \mathbf{y}_i - \mathbf{H} \left(\mathbf{x}_i^{ph}, \theta \right) \mathbf{g} - \mathbf{B} \left(\mathbf{x}_i^{pr}, \mathbf{x}_i^{ph} \right) \mathbf{d} \right\|_F^2 \\ + \lambda_1 \sum_{j=1}^N \left\| \mathbf{H} \left(\mathbf{x}_j^{ph}, \theta \right) \mathbf{g} - \varphi[\eta(\mathbf{x}_j^{ph}, \theta)] \right\|_F^2 \end{aligned}$$

$$\text{s.t. } \omega_L \leq \theta \leq \omega_U. \quad (8)$$

The problem in Eq. 8 is a sum of convex functions under a convex constraint and hence convex. We can utilize Algorithm 2 to solve it, with $M = 3$ and $f_1(\mathbf{g}) = \|\mathbf{Y} - \mathbf{H}_1 \mathbf{g} - \mathbf{B}_1 \mathbf{d}\|_F^2$, $f_2(\mathbf{g}) = \lambda_1 \|\mathbf{H}_2 \mathbf{g} - \tilde{\eta}\|_F^2$, $f_3(\mathbf{g}) = \sum_{k=1}^{q_3} \mathbf{I}_{\omega_{L_k} \leq \theta_k \leq \omega_{U_k}}$. The proximal operators of the functions that are given in Proposition 2. The derivation is given in Section B. in the supplementary materials.

Proposition 2: Let $f_1(\mathbf{g}) = \|\mathbf{Y} - \mathbf{H}_1 \mathbf{g} - \mathbf{B}_1 \mathbf{d}\|_F^2$, $f_2(\mathbf{g}) = \lambda_1 \|\mathbf{H}_2 \mathbf{g} - \tilde{\eta}\|_F^2$ and $f_3(\mathbf{g}) = \sum_{k=1}^{q_3} \mathbf{I}_{\omega_{L_k} \leq \theta_k \leq \omega_{U_k}}$. The proximal operators of $\Delta f_1, \dots, \Delta f_3$ are given as follows: Let $\mathbf{z}$ and $\mathbf{a}$ have the same size as θ .

- 1) If $\mathbf{z} = \text{prox}_{\Delta f_1}[\mathbf{a}]$, we have
- 2) $\text{prox}_{\Delta f_1}[\mathbf{a}] = (\mathbf{I} + \Delta \mathbf{g} \mathbf{g}^T)^{-1} (\mathbf{a} + \Delta \mathbf{g} (\mathbf{Y} - \mathbf{B}_1 \mathbf{d})^T)$, where $\mathbf{I}$ is the identity matrix of order p .
- 3) If $\mathbf{z} = \text{prox}_{\Delta f_2}[\mathbf{a}]$, we have
- 4) $\text{prox}_{\Delta f_2}[\mathbf{a}] = (\mathbf{I} + \lambda_1 \Delta \mathbf{g} \mathbf{g}^T)^{-1} (\mathbf{a} + \lambda_1 \Delta \mathbf{g} \tilde{\eta}^T)$, where $\mathbf{I}$ is the identity matrix of order p .
- 5) If $\mathbf{z} = \text{prox}_{\Delta f_3}[\mathbf{a}]$, we have

$$z_k = \text{prox}_{\Delta f_3}[a_k] = \max\{\omega_{L_k}, \min\{a_k, \omega_{U_k}\}\}.$$

3) *3. Identifiability and Uniqueness:* Another aspect of our algorithm that needs to be discussed is identifiability issues that may arise in evaluating terms on the form $\mathbf{H}_1 \mathbf{g}$ appearing in Eq. 7 for instance. This problem can be casted as a dictionary learning problem since we are in some sense trying to learn a basis matrix, $\mathbf{H}_1$, and a sparse coefficients vector, $\mathbf{g}$. The dictionary learning method does not assume that the patterns in $\mathbf{H}_1$ to be orthogonal like principal component analysis, and the reconstructed signal is often evaluated accurately under mild conditions. However, this does not guarantee that dictionary learning can uniquely recover sparse data from samples. In other words, the uniqueness of the learned basis and sparse coefficients vector is not always guaranteed, which may create an identifiability issue. The detailed statistical assumptions that guarantee the identifiability is stated in proposition 3 and further discussed in Section C. in the supplementary materials.

Proposition 3: Suppose $\mathbf{g}$ is a vector with at most $k \ll m$ nonzero entries, called k -sparse. Assume the basis $\mathbf{H}$ is spark on the vectors $\mathbf{g}$ such that:

$$\mathbf{H} \mathbf{g}_1 = \mathbf{H} \mathbf{g}_2 \text{ for } \mathbf{g}_1, \mathbf{g}_2 \in \mathbb{R}^m \Rightarrow \mathbf{g}_1 = \mathbf{g}_2$$

Assuming $\mathbf{H}$ is given, and our signal is constructed such that

$$\tilde{\mathbf{Y}} = \mathbf{H} \mathbf{g},$$

where $\tilde{\mathbf{Y}} = \mathbf{Y} - \mathbf{B}_1 \mathbf{d}$, a lower bound can be set on the number of samples, n , of $\tilde{\mathbf{Y}}$ that guarantees unique recovery of sparse vector $\mathbf{g}$ from a signal $\tilde{\mathbf{Y}}$ with a given basis matrix $\mathbf{A}$ such that:

$$n \geq Ck \log\left(\frac{m}{k}\right),$$

where C is a constant independent of m, n , and k .

In the case $\mathbf{H}$ is unknown, assuming the above holds, we assume we can subsample N similar samples of $\tilde{Y}$ from our system such that we have $\tilde{Y} = \{\tilde{Y}_1, \dots, \tilde{Y}_N\}$. Then, the uniqueness of both $\mathbf{g}$ and $\mathbf{H}$ can be ensured if we establish a lower bound on the required number of subsamples, N , for the uniqueness of $\mathbf{A}$ with a probability of $(1 - \beta)$ as follows:

$$N = \frac{k+1}{\beta} \binom{m}{k}.$$

The formula above implies that if the number of subsamples is sufficiently large, as specified in Proposition 3, the sparse matrix factorization of the basis matrix will be unique. For further discussion and proofs on the identifiability, readers are referred to [1].

IV. SIMULATION STUDIES

This section of the paper utilizes simulated data to explore and measure the capabilities and efficiency of our model. We have structured our simulation experiments, presented from sections IV to IV-B, with the aim of resolving the questions below:

- 1) How well does PWL perform under different sample sizes of the labeled data (Section IV)?
- 2) How does the accuracy of the physics-based model affect PWL performance (Section IV)?
- 3) How many labeled samples can be conserved through integrating samples generated from the physics-based model (Section IV-A)?

Data Generation: The process for generating data is uniform across all experiments, but the specific values of the parameters can vary. The process of generating simulated data begins with sampling the input variables $(\mathbf{x}_i^{pr}, \mathbf{x}_i^{ph})$ from a multivariate normal distribution, i.e., $(\mathbf{x}_i^{pr}, \mathbf{x}_i^{ph}) \sim MN(\mathbf{0}, \Sigma_x)$, where Σ_x is the covariance matrix. The calibration parameters are sampled from a uniform distribution, $\theta \in (0, 1]$. Then, we adopt the synthetic toy example from Bastos and O'Hagan [2] with some modification to simulate the output variable as follows: $\mathcal{Y}_i = \eta_{true}(\mathbf{x}_i^{ph}, \theta) + h(\mathbf{x}_i^{pr}) + \varepsilon_i$, where $\eta_{true}(\cdot, \cdot)$ is the physics-based model, $h(\cdot)$ is a function of the process variables not included in the physics-based model, and ε_i is sampled from $N(0, \sigma_\varepsilon^2)$. σ_ε^2 is the noise variance based on the selected signal-to-noise ratio $SNR = \frac{Var(\mathcal{Y})}{\sigma_\varepsilon^2}$. We assume in our simulation that we cannot observe $\eta_{true}(\cdot, \cdot)$ but a less accurate version of it as follows: $\eta(\mathbf{x}_i^{ph}, \theta) = \eta_{true}(\mathbf{x}_i^{ph}, \theta) + \delta(\mathbf{x}_i^{ph}) + \varepsilon_\eta$, where, $\delta(\cdot)$ is a function that simulates the discrepancy of the physics-based model, and ε_η is uncertainty in the physics-based model sampled from $N(0, \sigma_{\varepsilon_\eta}^2)$. In our simulations, the SNR is fixed at five, q_1 to three, q_2 to two, and q_3 to two. The functional forms for our models in the simulation are chosen as follows:

$$\eta_{true}(\mathbf{x}_i^{ph}, \theta) = \left(1 - \exp\left(-\frac{1}{2x_3}\right)\right) \times \frac{10\theta_1 x_1^3 + 19x_1 + 60}{9\theta_2 x_2^3 + 4x_2 + 20} \quad (9)$$

$$\delta(\mathbf{x}_i^{ph}) = \frac{10x_1^2 + 4x_2^2 + 2x_3^3}{50x_1x_2 + 10} \quad (10)$$

$$h(\mathbf{x}_i^{pr}) = \frac{4x_4^2 + 8x_5^2}{5x_4x_5} \quad (11)$$

Models Under Comparison: We compare PWL with a variety of supervised learning models in ML, which can only use labeled samples in training and cannot incorporate the physics-based model as PWL. The following models are compared against in the simulations:

- **Ridge regression (RR)** as representative of linear regression (LR) models, it uses quadratic loss and a squared L_2 regularization for the model's coefficients.
- **Support vector regression (SVR)** as representative of kernel learning methods. The kernel type selected is a polynomial kernel function which produced best results on our simulated dataset.
- **Decision tree (DT)** as representative of is a tree-based nonparametric supervised learning method. The maximum depth was selected based on prior research findings [4], [10] and best performance was selected in every experiment.
- **Random forests (RF)** as representative of ensemble methods and gradient boosted decision trees (GBDT) are typical choice in this case. For the hyperparameter settings, the number of boosting stages was selected from [30, 100, 160], learning rate was selected from [0.1, 0.15, 0.2], and the maximum depth was selected based on prior research findings [4], [6], [8], and the best performing parameters were selected in every experiment.
- **Gaussian process regression (GP)** as representative of nonparametric, Bayesian approaches. A squared exponential kernel showed best performance on our dataset.
- **Physics-based model (Physics)** as representative of physics-based models. The model is calibrated using a two-step procedure. First, calibration parameters were obtained through least-squares estimation as follows:

$$\hat{\theta} = \arg \min_{\theta \in [\omega_L, \omega_U]} \sum_{i=1}^n [\mathcal{Y}_i - f[\eta(\mathbf{x}_i^{ph}, \theta)]]^2 \quad (12)$$

Then, the model discrepancy was estimated using a gaussian process model similar to the approach described in Kennedy and O'Hagan [35] to achieve best performance.

Performance Metrics: To verify the effectiveness of the proposed method, three commonly used metrics in regression task are employed in this work, i.e., MSE, RMSE, and MAE. Besides, the mean values and the standard deviations are obtained by independent experiments with different random seeds to ensure the experiments reliability.

A. Model Performance With Varying Sample Sizes

The objective of this experiment is to assess and contrast the effectiveness of models using varying samples size. Twelve unique scenarios are constructed, beginning with a scenario of only 10 labeled samples, and progressively adding 10 more labeled samples in each subsequent scenario. This increment continues until the final scenario, which encompasses a total of 120 labeled samples. On the other hand, we assume in each scenario that there are 100 samples generated from the physics-based model $\eta(\cdot, \cdot)$. In each of these scenarios,

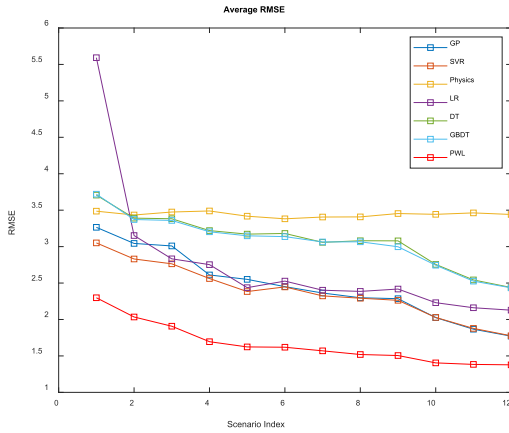


Fig. 2. Average test RMSE across all models.

we allocate 70% of the labeled samples for the training process and use the remaining 30% for validation. We also produce a separate test set with 200 labeled samples, which is employed to assess and compare the predictive performance across different models.

In each scenario, PWL is trained based on both the labeled samples and the samples generated from the physics-based model. For PWL, the three tuning parameters are determined by achieving the lowest MSE on the validation set. A similar approach is applied for tuning the parameters of other models, focusing on minimizing the MSE during validation. This training, validation, and testing process is repeated 20 times randomly for each model, with recoding both the average and standard deviation of their performance metrics. In Figure 2, it is evident that PWL consistently outperforms its counterparts in terms of average RMSE. To evaluate the statistical significance of this performance disparity, one-sided hypothesis tests are performed comparing PWL against each competing model. The results indicate that PWL's performance metrics are significantly better across all scenarios compared to the metrics of competing models.

Table I shows the detailed performance of three of our scenarios representing low, medium, and high labeled data availability in training. Moreover, the comparison between PWL and its competing models reveals that the largest performance gap is observed in cases with a limited number of labeled samples. This shows the advantage of PWL due to its capability of incorporating knowledge from the physics-based model to compensate for the shortage of labeled samples in training. Moreover, by comparing PWL with the physics-based model, we can observe that, regardless of increasing the number of training samples in training, the physics-based model exhibits a limit in its performance, due to the less flexible structure of the physics-based model.

Moreover, for a more comprehensive assessment that includes performance stability, Figure 3. features a boxplot analysis of the MSE in each scenario for PWL versus the best performance of the competing models. This analysis highlights a more rapid decrease in both the mean and the spread of MSE values for PWL. Additionally, the comparison reveals that the competing model exhibits greater fluctuations in both

TABLE I
ALL MODELS' PERFORMANCE METRICS ON TEST DATA FOR THREE SELECTED SCENARIOS

RMSE (Mean $\pm$ Std)						
Labeled samples in training						
Model	10	60	120			
LR	5.592 $\pm$ 3.751	2.527 $\pm$ 0.251	2.128 $\pm$ 0.143			
SVR	3.051 $\pm$ 0.478	2.447 $\pm$ 0.227	1.776 $\pm$ 0.127			
DT	3.707 $\pm$ 0.329	3.182 $\pm$ 0.310	2.439 $\pm$ 0.205			
GP	3.265 $\pm$ 0.497	2.451 $\pm$ 0.271	1.771 $\pm$ 0.129			
GBDT	3.713 $\pm$ 0.352	3.128 $\pm$ 0.323	2.423 $\pm$ 0.220			
Physics	3.242 $\pm$ 0.155	3.143 $\pm$ 0.161	3.201 $\pm$ 0.216			
PWL	2.298 $\pm$ 0.507	1.619 $\pm$ 0.200	1.378 $\pm$ 0.113			
MSE (Mean $\pm$ Std)						
Labeled samples in training						
Model	10	60	120			
LR	44.879 $\pm$ 79.772	6.445 $\pm$ 1.362	4.547 $\pm$ 0.613			
SVR	9.531 $\pm$ 3.034	6.037 $\pm$ 1.186	3.171 $\pm$ 0.453			
DT	13.844 $\pm$ 2.420	10.221 $\pm$ 2.033	5.990 $\pm$ 0.999			
GP	10.898 $\pm$ 3.254	6.078 $\pm$ 1.412	3.153 $\pm$ 0.457			
GBDT	13.904 $\pm$ 2.617	9.882 $\pm$ 2.089	5.919 $\pm$ 1.072			
Physics	10.536 $\pm$ 1.001	9.905 $\pm$ 1.017	10.293 $\pm$ 1.373			
PWL	5.530 $\pm$ 2.696	2.660 $\pm$ 0.710	1.910 $\pm$ 0.314			
MAE (Mean $\pm$ Std)						
Labeled samples in training						
Model	10	60	120			
LR	4.492 $\pm$ 3.087	1.958 $\pm$ 0.193	1.659 $\pm$ 0.100			
SVR	2.413 $\pm$ 0.372	1.924 $\pm$ 0.175	1.420 $\pm$ 0.109			
DT	2.959 $\pm$ 0.250	2.496 $\pm$ 0.252	1.901 $\pm$ 0.149			
GP	2.572 $\pm$ 0.380	1.912 $\pm$ 0.204	1.420 $\pm$ 0.110			
GBDT	2.963 $\pm$ 0.262	2.463 $\pm$ 0.251	1.898 $\pm$ 0.167			
Physics	2.750 $\pm$ 0.156	2.650 $\pm$ 0.150	2.719 $\pm$ 0.202			
PWL	1.821 $\pm$ 0.405	1.284 $\pm$ 0.159	1.088 $\pm$ 0.094			

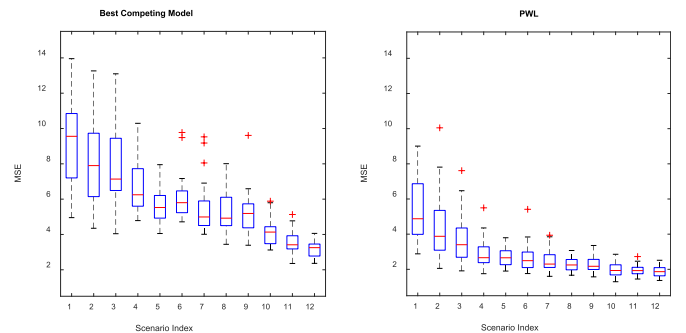


Fig. 3. Boxplots of MSE for PWL vs. best competing model.

variance and mean, suggesting that its performance is less robust compared to PWL.

B. Model Performance Under Different Physics-Based Model Accuracies

The findings in the previous section showed that PWL can help improve performance under a limited number of labeled samples compared with competing models. The improvement is achieved by incorporating physical knowledge in training, which can help mitigate the shortage of labeled samples.

TABLE II
PWL PERFORMANCE UNDER DIFFERENT PHYSICS-BASED MODEL ACCURACIES VS. COMPETING MODELS

Model	RMSE (Mean $\pm$ Std)		MSE (Mean $\pm$ Std)		MAE (Mean $\pm$ Std)	
LR	2.409	$\pm$ 0.319	5.900	$\pm$ 1.626	1.882	$\pm$ 0.260
SVR	2.311	$\pm$ 0.296	5.422	$\pm$ 1.511	1.831	$\pm$ 0.257
DT	3.042	$\pm$ 0.315	9.347	$\pm$ 2.044	2.414	$\pm$ 0.265
GP	2.336	$\pm$ 0.214	5.501	$\pm$ 1.043	1.829	$\pm$ 0.184
GBDT	3.068	$\pm$ 0.384	9.555	$\pm$ 2.480	2.443	$\pm$ 0.314
PWL-H	1.392	$\pm$ 0.161	1.962	$\pm$ 0.462	1.106	$\pm$ 0.129
PWL-M	2.054	$\pm$ 0.251	4.279	$\pm$ 1.056	1.614	$\pm$ 0.198
PWL-L	2.404	$\pm$ 0.279	5.854	$\pm$ 1.343	1.890	$\pm$ 0.233

This result leads to the question regarding how much the performance will be affected by the inaccuracy of the physics-based model.

To answer this question, we focus in this experiment on the physics-based model $\eta(\cdot, \cdot)$ used in PWL. We use Pearson correlation between the label y_i and the output of the physics-based model $\eta(\cdot, \cdot)$ as a metric for the accuracy of the physics-based model. The correlation will be controlled through varying the magnitude of the discrepancy function $\delta(x_i^{ph})$. We create three different scenarios over which the physics-based model accuracy will vary, including a highly accurate model (i.e., scenario 1) with a correlation of 0.85, a mildly accurate model (i.e., scenario 2) with a correlation of 0.7, and an insufficiently accurate model (i.e., scenario 3) with a correlation of 0.5. All scenarios are based on 40 labeled samples and 100 samples generated from the physics-based model $\eta(\cdot, \cdot)$. Table II shows the performance metrics of the experiments. We can see that the accuracy of the physics-based model significantly affects PWL performance with the best performance achieved with the highly accurate physics-based model (i.e., PWL-H). However, one important observation is that the performance of PWL with insufficiently accurate physics-based model (i.e., PWL-L) is statistically undisguisable from the best performance achieved by competing methods. This indicates that even when the physics-based model is not sufficiently accurate, we can expect that the performance of PWL is still comparable to competing methods that only include labeled samples in training. This result is due to the fact that PWL utilizes a tuning λ_1 to control how much the physics-based model should be emphasized in the training process.

C. Saving of Labeled Samples Using PWL

The previous experiments showed that PWL can be very helpful when the labeled samples are limited, compared with models that rely only on the labeled samples. Hence, it is desirable to learn how many labeled samples can be saved by adopting PWL in order to achieve the same performance as other models. To answer this question, we focus on scenario 3 in Section IV-A, in which PWL achieved an average MSE of 3.745 with a training set of 30 labeled samples and

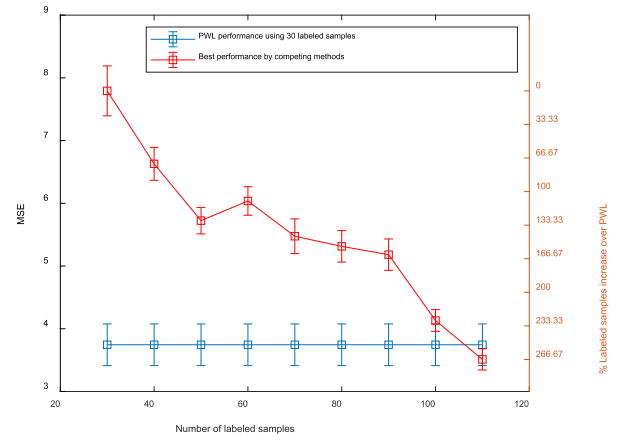


Fig. 4. MSE performance of competing models with increasing labeled sample size.

100 samples from the physics-based model. To determine how many extra labeled samples are required for the competing models to achieve the same MSE as PWL, the number of labeled samples is varied incrementally from 30 to 110 with 10 sample increments. Under each sample size, all competing models are trained and only the top-performing model's results are record. Figure 4 shows the MSE of best competing model (red curve) in comparison with PWL (blue curve).

We can observe that the MSE of competing models drops to be lower than that of PWL with 110 labeled samples in training. We conduct one-sided hypothesis testing to see when the performance of competing models has statistically equivalent MSE to PWL. The result shows that statistical equivalence is achieved when 100 samples are included in training. This indicates that an increment of 233.3% in labeled samples is required by the competing models to achieve the same performance of PWL. Hence, the benefit of PWL in situations of costly data collection is substantial.

V. CASE STUDIES

In this section, we will present the application of PWL in two real datasets.

A. Temperature Difference Prediction in Spot Welding

In spot welding, high temperature difference usually results in a poor-quality weld. The objective of this study is to accurately predict temperature increase using process variables.

To validate our proposed PWL framework, we conducted extensive spot-welding experiments using an Electric X-type Welding Gun with a 75-kVA capacity and 50Hz frequency. During welding, in-line measurements of voltage, current, and pressure variations were recorded. The nugget formation and temperature of the welding zone were monitored using an infrared camera (TACHYON 1024 μ CAMERA). Spherical tip electrodes made of CuCrZr alloy conforming to DIN 44750, Part 1, were used, and the electrode tips were replaced every 5 welds to avoid alloying and eliminate potential deformations. The electrodes had a ball radius of 75mm and a diameter of 20mm. For the workpieces, we used uncoated steel plates commonly used in the automotive industry, specifically low

carbon EED quality steel (St1203 according to DIN1623), cut into 20×40 -mm pieces with a thickness varying between 1 and 3mm. To prepare the specimens, the front face of each welded pair was ground flat with a file and finished with a 600x abrasive.

There are four controllable variables in the welding process: force [kN] (x_1), current [kA] (x_2), time of weld [s] (x_3), mass [g] (x_4), and the objective is to predict the temperature difference on a part $[\Delta T](y)$ given input setting. The time of weld represents the length of time during which the welding process is actively conducted on the workpiece. Process data are obtained from 35 welding experiments we conducted using a Latin Hypercube design over the range of controllable parameters. The physics-based model $\eta(.,.)$ is represented by a mathematical equation that has a set of input variables that are different from the experiment as follows:

$$\Delta T = \frac{I^2 \cdot \rho \cdot L \cdot t}{A \cdot m \cdot C_p}, \quad (13)$$

where

- I: current [kA],
- t: time of weld [s],
- m: mass [g],
- ρ : Resistivity [ohm/m],
- C_p : thermal conductivity [W/mK],
- L: Length of path [m],
- A: Cross sectional area [m].

We can observe that the physics-based model includes three of the controllable variables from our process setting (i.e., x_2, x_3, x_4), which we refer to in PWL as $\mathbf{x}^{ph}$, and the variables that are only included in the process setting (i.e., x_1) are referred to as $\mathbf{x}^{pr}$. On the other hand, there are some parameters that are only included in the physics-based model, which PWL refers to as θ . We calibrate the physics-based model to our process samples according to the two-step procedure we described in Eq. 12. The physics-based model shows a good correlation with our process samples of 0.884, however a poor predictive performance with RMSE of 72.8. This type of physics-based model represents a desirable candidate for our PWL framework. In the first experiment, we will compare PWL performance with the benchmarking methods described in Section IV. We will use the 35 available samples and 35 samples from the physics equation using the same variable setting in process, and report the 5-fold cross validation mean and standard deviation performance of RMSE, MSE, and MAE from 5 random runs. Table III shows the performance recorded in our experiment. PWL performance in all metrics is superior to competing models. PWL not only has the smallest mean in all metrics, but also shows good robustness with small standard deviation compared to competing models.

To assess the impact of excluding variables such as force, we can compare the performance of the standalone physics-based model against that of the comprehensive PWL model, as shown in Table III. This comparison not only underscores the limitations of the physics-based model when used in isolation but also demonstrates the improved accuracy and reliability achieved through its seamless integration with

TABLE III
PWL PERFORMANCE VS. COMPETING MODELS ON CASE STUDY A

Model	RMSE (Mean $\pm$ Std)		MSE (Mean $\pm$ Std)		MAE (Mean $\pm$ Std)	
LR	32.68	$\pm$ 10.84	1177.49	$\pm$ 839.79	26.62	$\pm$ 9.78
SVR	34.04	$\pm$ 14.43	1352.99	$\pm$ 1141.79	27.69	$\pm$ 12.92
DT	68.11	$\pm$ 29.41	5359.43	$\pm$ 5260.32	56.46	$\pm$ 22.79
GP	24.88	$\pm$ 8.93	693.46	$\pm$ 466.24	21.37	$\pm$ 7.82
GBDT	62.80	$\pm$ 23.22	4393.36	$\pm$ 3545.8	50.56	$\pm$ 17.59
Physics	73.62	$\pm$ 3.74	5432.69	$\pm$ 561.04	61.95	$\pm$ 4.21
PWL	20.75	$\pm$ 8.34	495.49	$\pm$ 331.18	16.99	$\pm$ 7.76

TABLE IV
PWL PERFORMANCE VS. SEMI-SUPERVISED LEARNING MODELS ON CASE STUDY A

Model	RMSE (Mean $\pm$ Std)		MSE (Mean $\pm$ Std)		MAE (Mean $\pm$ Std)	
COREG	40.6	$\pm$ 8.0	1695.6	$\pm$ 667.0	31.7	$\pm$ 5.2
ICT	34.4	$\pm$ 13.3	1280.7	$\pm$ 1086.1	35.0	$\pm$ 7.7
Π Model	32.5	$\pm$ 12.8	1093.8	$\pm$ 1011.2	31.6	$\pm$ 8.6
Mean Teacher	32.9	$\pm$ 12.3	1118.1	$\pm$ 960.8	32.1	$\pm$ 7.4
PWL	26.6	$\pm$ 5.2	730.7	$\pm$ 278.0	21.8	$\pm$ 5.3

machine learning. Such an analysis provides valuable insights into the roles of missing variables in the physics-based model and the overall effectiveness of our PWL approach, thereby offering a more holistic understanding of temperature differences in spot welding.

Another aspect of PWL that we want to explore particularly in this case study is the performance under limited labeled data availability. The number of labeled samples provided in this case study is 35, which is considered a small number to fully understand predictors-response relationship. This phenomenon is very common especially in experiments that require the sample to be destroyed in some process setting to get a response reading. Hence, a model such as PWL that requires only a few labeled samples and is able to exploit unlabeled samples is highly desirable in this setting. Therefore, we will compare PWL performance with models that can utilize labeled and unlabeled samples in training such as semi-supervised learning models. The models to be compared with are CoReg [58], ICT [54], Π Model [36], and Mean Teacher [53]. Among the 35 labeled samples, we will consider only 15 to be labeled and the rest as unlabeled. We conducted 5 random runs and the performance on the unlabeled samples is reported in Table IV.

We can observe in Table IV that PWL performance is superior in mean RMSE, MSE and MAE over all competing models. PWL also shows a smaller standard deviation in all of the three metrics in Table IV, which indicates a more stable performance. Based on the observations in Tables III and IV, semi-supervised models demonstrated statistically significant superior performance compared to certain fully supervised models, particularly GBDT and DT. This outcome

TABLE V
PWL PERFORMANCE VS. COMPETING MODELS ON CASE STUDY B

RMSE (Mean $\pm$ Std)					
Labeled samples in training					
Model	10		100		
LR	0.5835	$\pm$	0.0759	0.4629	$\pm$ 0.0469
SVR	0.6075	$\pm$	0.0526	0.4590	$\pm$ 0.0518
DT	1.0664	$\pm$	0.0371	0.4823	$\pm$ 0.0904
GP	0.5981	$\pm$	0.0784	0.4708	$\pm$ 0.0758
GBDT	1.1420	$\pm$	0.0654	0.5872	$\pm$ 0.0576
Physics	0.9021	$\pm$	0.0395	0.8279	$\pm$ 0.1138
PWL	0.5566	$\pm$	0.0328	0.4519	$\pm$ 0.0497
MSE (Mean $\pm$ Std)					
Labeled samples in training					
Model	10		100		
LR	0.3443	$\pm$	0.0907	0.2158	$\pm$ 0.0444
SVR	0.3709	$\pm$	0.0639	0.2124	$\pm$ 0.0486
DT	1.1382	$\pm$	0.0795	0.2380	$\pm$ 0.0844
GP	0.3618	$\pm$	0.0913	0.2255	$\pm$ 0.0681
GBDT	1.3069	$\pm$	0.1510	0.3470	$\pm$ 0.0671
Physics	0.8149	$\pm$	0.0711	0.6940	$\pm$ 0.1842
PWL	0.3105	$\pm$	0.0361	0.2060	$\pm$ 0.0458
MAE (Mean $\pm$ Std)					
Labeled samples in training					
Model	10		100		
LR	0.4769	$\pm$	0.0612	0.3696	$\pm$ 0.0311
SVR	0.4960	$\pm$	0.0472	0.3643	$\pm$ 0.0354
DT	0.8842	$\pm$	0.0190	0.3863	$\pm$ 0.0615
GP	0.4852	$\pm$	0.0612	0.3746	$\pm$ 0.0410
GBDT	0.9354	$\pm$	0.0461	0.4821	$\pm$ 0.0571
Physics	0.7582	$\pm$	0.0485	0.7055	$\pm$ 0.0989
PWL	0.4578	$\pm$	0.0293	0.3534	$\pm$ 0.0194

is often observed in scenarios with limited data availability. Semi-supervised models excel by utilizing both labeled and unlabeled data, thereby providing a more holistic understanding of the dataset than fully supervised models, which rely exclusively on labeled data. In contrast, fully supervised models such as GBDT and DT are more prone to overfitting when faced with a data-limited situations.

B. Nugget Diameter Prediction in Spot Welding

In this section, we consider the spot-welding experiment given in [3] and [27]. There are three variables in this

TABLE VI
PWL PERFORMANCE VS. SEMI-SUPERVISED LEARNING MODELS ON CASE STUDY B

Labeled Samples		10					
Model		RMSE (Mean $\pm$ Std)		MSE (Mean $\pm$ Std)		MAE (Mean $\pm$ Std)	
COREG		0.5843	$\pm$ 0.0713	0.3464	$\pm$ 0.0852	0.4671	$\pm$ 0.0595
ICT		0.5892	$\pm$ 0.0672	0.3517	$\pm$ 0.0804	0.4740	$\pm$ 0.0559
Π Model		0.5794	$\pm$ 0.0679	0.3404	$\pm$ 0.0787	0.4681	$\pm$ 0.0616
Mean Teacher		0.5893	$\pm$ 0.0721	0.3525	$\pm$ 0.0859	0.4691	$\pm$ 0.0568
PWL		0.5566	$\pm$ 0.0328	0.3105	$\pm$ 0.0361	0.4578	$\pm$ 0.0293
Labeled Samples		100					
Model		RMSE (Mean $\pm$ Std)		MSE (Mean $\pm$ Std)		MAE (Mean $\pm$ Std)	
COREG		0.5024	$\pm$ 0.0319	0.2533	$\pm$ 0.0317	0.4147	$\pm$ 0.0452
ICT		0.5089	$\pm$ 0.0469	0.2612	$\pm$ 0.0482	0.4138	$\pm$ 0.0556
Π Model		0.5126	$\pm$ 0.0353	0.2640	$\pm$ 0.0361	0.4202	$\pm$ 0.0299
Mean Teacher		0.5491	$\pm$ 0.0395	0.3031	$\pm$ 0.0443	0.4592	$\pm$ 0.0455
PWL		0.4519	$\pm$ 0.0497	0.2060	$\pm$ 0.0458	0.3534	$\pm$ 0.0194

process: load (x_1), current (x_2), and gage (x_3), and the objective is to optimize the nugget diameter of the weld (y). The physics-based model is quite complex and contains one calibration parameter (θ) additional to the three variables in the process. The physics-based model uses coupled partial differential equations to represent thermal, electrical, and mechanical interactions, implemented via the ANSYS commercial code, as detailed in [3].

The first step is to approximate the physics-based model. Because there is no random error in the physics-based computer simulation model, an interpolating model such as kriging is suitable for the model approximation [49]. We fit an ordinary kriging model with Gaussian correlation function using the 35 simulations in Table III of [3]. Afterwards, the produced physics-based model is calibrated to the process samples described in Table IV of [3] according to the two-step procedure we described in Eq. 12. The process samples in [3] represent 12 experimental settings with 10 replicates per setting. In this case study, we obtained more process samples, 120 in total, than the previous case study. Thus, we are able to test PWL performance under low and high data availability scenarios. We will compare two scenarios, one with only 10 labeled process samples in training and the other with 100 process samples in training. The remaining samples are used for testing, and we will conduct five random splits to report the mean and the standard deviation of RMSE, MSE, and MAE.

Table V shows PWL performance compared to benchmarks under the two data availability scenarios. Generally, the best MSE value should be the closest to the variance of noise in data which was estimated to be 0.2 from the replicates. PWL achieved this desired MSE performance with a value of 0.206 under the scenario with 100 labeled samples in training. We can also observe that PWL outperforms the benchmarking methods in all performance metrics under the scenario of limited data availability.

In the results in Table VI, the PWL model consistently outperforms other models for both 10 and 100 labeled samples,

showcasing superior average performance and greater consistency across different runs as indicated by its low standard deviations. Among the semi-supervised models, their performances are relatively close, especially with a smaller labeled dataset which suggests comparable benefits in their strategies of leveraging unlabeled data. Furthermore, the performance of PWL improves more significantly with the increase in labeled samples compared to the other methods. This indicates that PWL is effective at utilizing unlabeled data and scales well with more labeled data.

VI. CONCLUSION

In this paper, we introduce a novel methodology to enhance ML model performance in manufacturing processes utilizing the weak supervision paradigm. Traditional methods relying on expensive, time-consuming labeled datasets have limitations, which we overcome by integrating partial physical knowledge, significantly improving model performance. Our approach is validated by comprehensive simulations and two case studies, demonstrating robustness across various noise levels and numbers of labeled samples, while also highlighting the immense potential of our proposed method in diverse manufacturing processes.

Methodologically, our approach offers several advantages, including substantial improvements in predictive performance and efficient optimization via the ADMM algorithm. This paper establishes that weak supervision is a promising alternative to conventional approaches for enhancing the accuracy of ML models in manufacturing processes, especially when labeled datasets are either unavailable or prohibitively expensive to acquire.

In summary, our methodology holds the potential to reduce costs, increase efficiency, and improve in-process quality in manufacturing processes. The findings of this study open up exciting new directions for future research. A direct extension could involve adapting our flexible PWL model to manage more complex or higher-dimensional outputs, while addressing the new challenges that such outputs may present. Furthermore, our framework could be expanded to tackle intrinsic uncertainties in some physics-based models. For instances where the output variable is within a known range at a given confidence, the PWL framework might be adapted by integrating techniques like weighted loss functions, soft constraints on predictions, and probabilistic outputs, though these enhancements may add complexity to optimization.

Researchers can also explore our approach's applicability in other fields with scarce labeled data, such as non-destructive testing or predictive maintenance. There is also potential for integrating our framework with advanced machine learning techniques such as deep learning and reinforcement learning to improve predictive performance; or with weak supervision methods like crowdsourcing or semi-supervised learning to build robust models with limited labeled data. Lastly, examining the impact of incorporating domain-specific knowledge, such as expert heuristics or insights from materials science, could further improve the method's effectiveness in the manufacturing domain.

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